

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Computer Science and Engineering) SEMESTER - 3 Winter 2025 (Regular)

Course :Bachelor of Technology (Computer Science and Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: 25AF1245PC303 - DISCRETE MATHEMATICS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

1. Power set of empty set has exactlySubsets.
A. One B. Two C. Zero D. Three
2. A binary operation Δ on a set of integers is defined as $x\Delta y = x + y$ which one of the following statement is true about?
A. commutative but not associative B. Both commutative and associative
C. associative but not commutative D. neither commutative nor associative
3. Let A be finite set having x elements and B be a finite set having y elements. What is the number of distinct functions mapping B into A.
A. x B. 2 C. y D. $y!/(y-x)!$
4. Which of the following is NOT a proposition?
A. 7 is a prime number B. $x + 3 = 10$
C. Mumbai is the capital of Maharashtra D. $2 + 2 = 4$
5. Which of the following is logically equivalent to $\neg(P \wedge Q)$?
A. $\neg P \wedge \neg Q$ B. $\neg P \vee \neg Q$ C. $P \vee Q$ D. $\neg P \rightarrow \neg Q$
6. The negation of the statement $\forall x(P(x))$ is:
A. $\forall x(\neg P(x))$ B. $\exists x(\neg P(x))$ C. $\neg \exists x(P(x))$ D. $\exists x(P(x))$
7. Which of the following statements is NOT equivalent to $\forall x \exists y P(x, y)$?
A. For every x, there exists a y such that $P(x, y)$
B. $\exists y \forall x P(x, y)$
C. For each x, some y satisfies $P(x, y)$
D. Every x has at least one y with property P
8. Which of the following structures must have an identity and inverses?
A. Semigroup B. Monoid C. Group D. Ring
9. Which algorithm is used to find a minimum spanning tree?
A. Breadth First Search B. Depth First Search
C. Dijkstra's Algorithm D. Kruskal's Algorithm

- 10 A connected graph has an Euler circuit if:
 A. All vertices have odd degree
 B. Exactly two vertices have odd degree
 C. All vertices have even degree
 D. The graph is complete
- 11 A graph that has no cycles is called:
 A. Complete graph B. Bipartite graph C. Tree D. Planar graph
- 12 Which rule of inference is used in the argument: If $P \rightarrow Q$ and P is true, then Q is true.
 A. Modus Tollens B. Hypothetical Syllogism
 C. Modus Ponens D. Resolution

Q2. Solve the following.

- A) Translate each of these statements into logical expressions using predicates, 6 quantifiers, and logical connectives. Let $T(x)$ = x is a tool, $C(x)$ = x is in the correct place, and $E(x)$ = x is in excellent condition.
 a) All tools are in the correct place and are in excellent condition.
 b) Nothing is in the correct place and is in excellent condition.
 c) One of your tools is not in correct place, but it is in excellent condition.
- B) Show that the argument form with premises $(p \wedge t) \rightarrow (r \vee s)$, $q \rightarrow (u \wedge t)$, $u \rightarrow 6$ p , and $\neg s$ and conclusion $q \rightarrow r$ is valid by using rules of inference.

Q3. Solve the following.

- A) Show that $A + B = (A \cup B) \cap (\sim A \cup \sim B)$ 6
- B) Given $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$. Let xRy whenever y is divisible by x . Find and 6 Write R as set of ordered pairs. Is R so found reflexive, symmetric and transitive? Is R Equivalence? Justify with a valid argument.

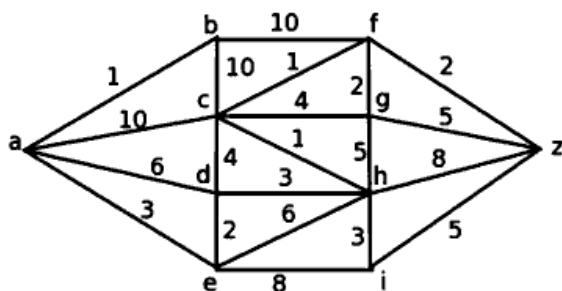
Q4. Solve Any Two of the following.

- A) It is given that white tiger population of Orissa (INDIA) is 30 at time $n = 0$ and 326 at time $n = 1$. Also the increase from time $n - 1$ to time n is twice the increase from time $n - 2$ to $n - 1$. Write and solve recurrence relation for growth rate of tiger.
- B) For the difference equation given below find particular solution. 6

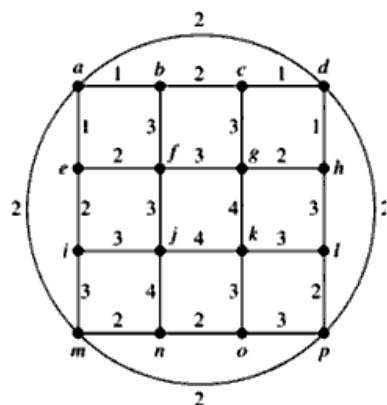
$$a_r - 4a_{r-1} + 4a_{r-2} = (r + 1)2^r$$
- C) Solve the following non-homogeneous recurrence relation $a_n = 2a_{n-1} + 3(2^n)$ 6 where, $a_0 = 3$.

Q5. Solve Any Two of the following.

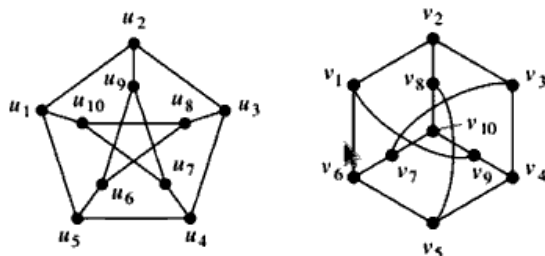
- A) Apply the Dijkstra's Algorithm to find the length of a shortest path between **a** and **6** **z** in the given weighted graph.



- B) Use Kruskal's algorithm to find a minimum spanning tree for the weighted graph in 6 figure given below. 6



- C) With justification determine whether the given pair of graphs is isomorphic or not. 6



Q6. Solve Any Two of the following.

- A) Let G be a reduced residue system modulo 15, say, $G = \{1, 2, 4, 7, 8, 11, 13, 14\}$. 6
 (i) Find Multiplication table of G .
 (ii) Find 2^{-1} , 7^{-1} .
 (iii) Find the orders and subgroups generated by 2, 7, and 11.
- B) Define the following Terms: 6
 (i) Abelian Group (ii) Identity Element (iii) Inverse Element
 (iv) Group (v) Monoid (vi) Rings
- C) Consider the set N of positive integers, and let $*$ denote the operation on least 6 common multiple (lcm) on N .
 (i) Find $4 * 6$ and $3 * 5$.
 (ii) Is $(N, *)$ is a semigroup? Is it commutative?
 (iii) Which element in N , if any, have inverse and what are they?

*** End ***